

Laboratory Work 3

Merged Data, Star Schema & DAX Function Insights

Submitted by: **Edmundo C. Ederio Jr.**

Instructor: **Joseph A. Vistal**

Course: **IS 107 – Business Intelligence**

PART 1

Data Modeling

Dataset: Merged_Sales_Data • Objective: Build a Star Schema

Part 1 – Review the Merged Dataset

MessySales	OrderID	CustomerID	OrderDate	SalesAmount	CustomerName	
Customers	1 O001	C009	1/4/2025	3000	Customer 9	C
Orders	2 O109	C009	3/7/2025	2200	Customer 9	C
	3 O112	C009	1/20/2025	3000	Customer 9	C
	4 O188	C009	3/18/2025	7500	Customer 9	C
	5 O002	C156	2/12/2025	7500	Customer 156	D
	6 O020	C156	1/28/2025	800	Customer 156	D
	7 O130	C156	3/9/2025	45000	Customer 156	D
	8 O173	C001	2/6/2025	3000	Customer 1	L
	9 O003	C137	3/11/2025	800	Customer 137	D
	10 O032	C137	3/21/2025	1200	Customer 137	D
	11 O080	C002	2/2/2025	800	Customer 2	M
	12 O167	C002	1/4/2025	2200	Customer 2	M
	13 O004	C127	1/1/2025	45000	Customer 127	T
	14 O028	C127	1/10/2025	800	Customer 127	T
	15 O148	C003	1/17/2025	1200	Customer 3	B
	16 O005	C105	3/3/2025	1200	Customer 105	B
	17 O179	C004	1/8/2025	1200	Customer 4	D
	18 O006	C054	3/29/2025	2200	Customer 54	C
	19 O045	C054	3/30/2025	1200	Customer 54	C

Guide Questions

1. Is the dataset in flat-table format?→ Yes, there is 7 columns.
2. Which columns describe customers?→ CustomerName, City, Region
3. Which columns are transactions? → OrderID, OrderDate, SalesAmount

Part 2 – Identify Fact and Dimension Tables

CREATE DIMCUSTOMER TABLE

Queries [4]

- MessySales
- Customers
- Orders
- DimCustomer

Formula Bar: `= Table.Distinct(#"Removed Other Columns", {"CustomerID"})`

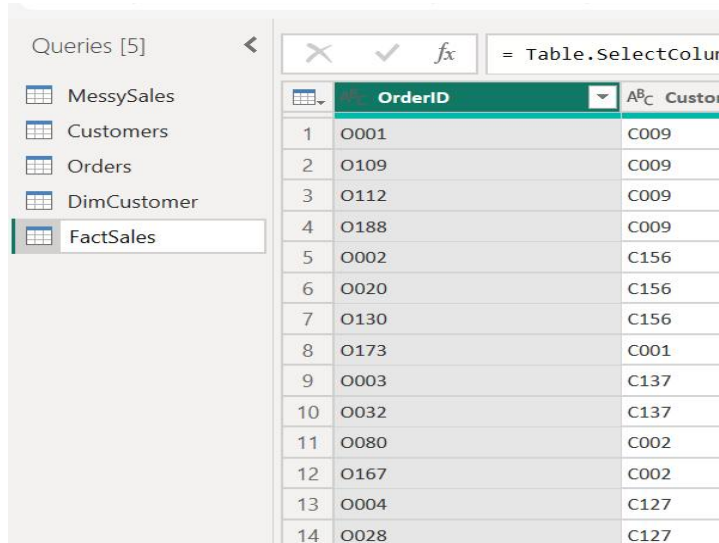
	CustomerID	CustomerName	City	Region
1	C009	Customer 9	Cagayan de Oro	Mindanao
2	C156	Customer 156	Dumaguete	Visayas
3	C001	Customer 1	Legazpi	Luzon
4	C137	Customer 137	Davao	Mindanao
5	C002	Customer 2	Manila	Luzon
6	C127	Customer 127	Tacloban	Visayas
7	C003	Customer 3	Butuan	Mindanao
8	C105	Customer 105	Butuan	Mindanao
9	C004	Customer 4	Dumaguete	Visayas
10	C054	Customer 54	Cagayan de Oro	Mindanao
11	C091	Customer 91	Manila	Luzon

STEPS

- (1) Right-click Orders → Reference
- (2) Choose Columns → keep CustomerID, CustomerName, City, Region
- (3) Remove Duplicates on CustomerID
- (4) Rename to DimCustomer

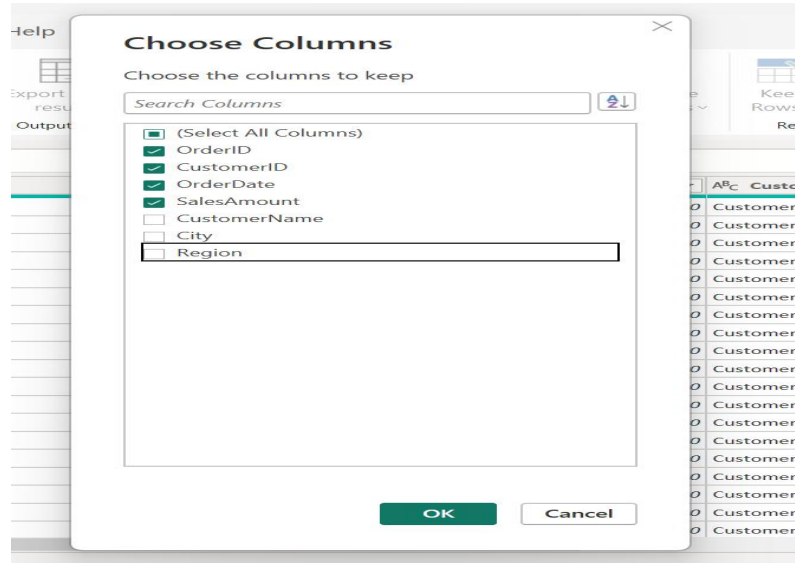
Part 2 – Identify Fact and Dimension Tables

CREATE FACTSALES TABLE



The screenshot shows the Power BI Desktop interface. On the left, the 'Queries [5]' pane lists several tables: MessySales, Customers, Orders, DimCustomer, and FactSales. The FactSales table is selected. The main view displays the data for FactSales, which is a table with three columns: OrderID, CustomerID, and SalesAmount. The data is as follows:

OrderID	CustomerID	SalesAmount
1	O001	C009
2	O109	C009
3	O112	C009
4	O188	C009
5	O002	C156
6	O020	C156
7	O130	C156
8	O173	C001
9	O003	C137
10	O032	C137
11	O080	C002
12	O167	C002
13	O004	C127
14	O028	C127



1. Why must dimension tables contain unique keys?

There must be unique so Power BI can create a valid one-to-many relationship since duplicate keys can cause incorrect data aggregation.

2. What problems occur if duplicates exist in DimCustomer?

Filters from DimCustomer would apply incorrectly to FactSales, producing inflated or wrong totals.

Part 3 – Define Primary & Foreign Keys

Table	Column	Key Type
DimCustomer	CustomerID	Primary Key (PK)
FactSales	CustomerID	Foreign Key (FK)
FactSales	OrderID	Primary Key (PK)

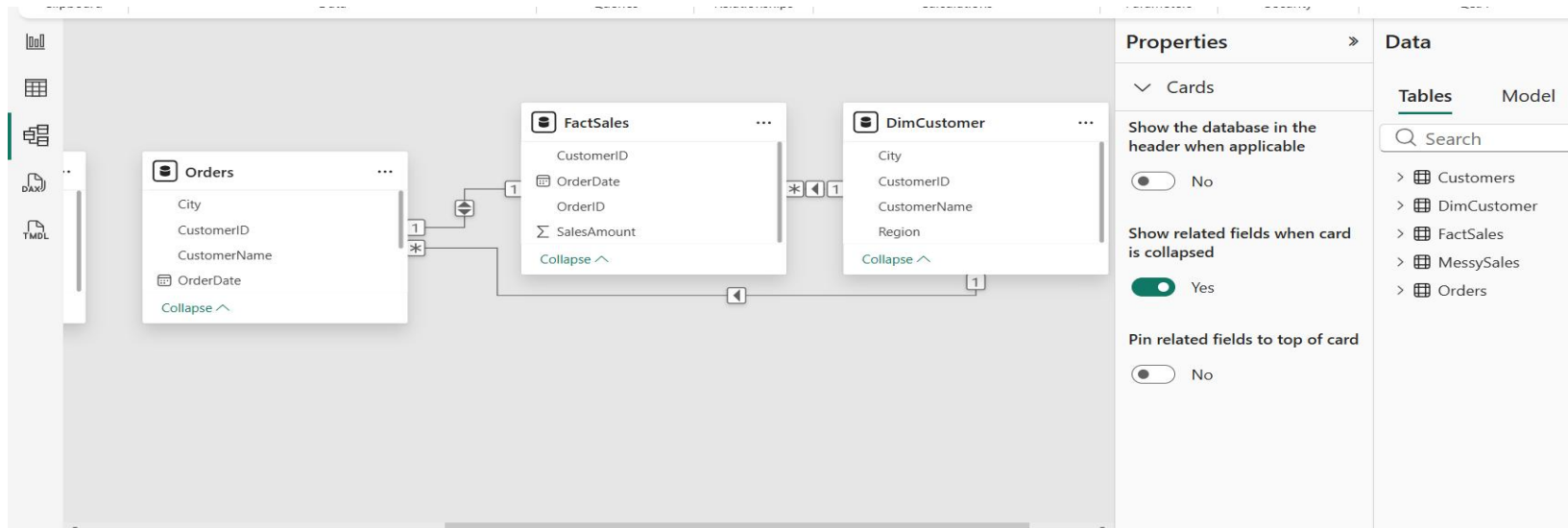
1. What is a primary key?

A Primary Key (PK) is a column with unique, non-null values that identifies each row.

2. Why is CustomerID a foreign key in FactSales?

The CustomerID in the FactSales table serves as a foreign key that links to the primary key in DimCustomer. This relationship enables Power BI to filter and analyze sales data using customer attributes like Region and City.

Part 4 – Create Relationships in Model View



1. Why should relationships flow from dimension to fact?

Filters propagate from the dimension table (one side) to the fact table (many side), allowing FactSales to be sliced by attributes like customer or date without causing circular filtering problems.

2. What happens if the relationship is Many-to-Many?

Power BI cannot resolve ambiguous joins — results become unreliable, totals may inflate, and cross-filtering becomes unpredictable.

Part 5 – Validate the Data Model

The screenshot displays the Microsoft Power BI Desktop interface. The ribbon menu at the top includes the following tabs: Clipboard (Paste, Copy, Format painter), Data (Get data, Excel workbook, OneLake catalog, SQL Server, Enter data, Dataverse, Recent sources), Queries (Transform data, Refresh), Visuals (New visual, Text box, More visuals), Calculations (New visual calculation, New measure, Quick measure), and Sensitivity (Sensitivity). The main workspace shows a table visual with the following data:

CustomerName	Region	Sum of SalesAmount
Customer 1	Luzon	5200
Customer 10	Mindanao	1200
Customer 100	Mindanao	16500
Customer 101	Mindanao	6000
Customer 102	Luzon	90800
Customer 103	Mindanao	4200
Customer 104	Visayas	3700
Customer 105	Mindanao	1200
Customer 106	Visayas	52500
Customer 107	Mindanao	9000
Customer 108	Mindanao	46200
Customer 11	Visayas	1200
Customer 110	Visayas	47200
Customer 111	Mindanao	800
Customer 112	Mindanao	5400
Customer 113	Luzon	91500
Customer 114	Mindanao	5200
Customer 115	Mindanao	46200
Customer 116	Luzon	49200
Customer 118	Mindanao	49500
Customer 119	Visayas	46600
Customer 12	Mindanao	12400
Customer 121	Luzon	2200
Total		3894400

On the right side, the 'Filters' pane is visible with a search bar and two sections: 'Filters on this page' and 'Filters on all pages', each containing an 'Add data fields here' button. The 'Visualizations' pane is also visible, showing various chart types and a 'Build visual' section with a search bar and a 'Values' section with an 'Add data fields here' button.

1. Did sales aggregate correctly per customer?

Yes. The table visual displays CustomerName, Region, and the total SalesAmount, with a Grand Total of 3,894,400. This indicates that the relationship between FactSales and DimCustomer is functioning properly.

2. What would happen if relationships were missing?

CustomerName and Region would appear as blank fields, and slicers or cross-filtering between tables would no longer function properly.

Enhancement Activities

From table

FactSales

CustomerID	OrderDate	OrderID	SalesAmount
C009	Saturday, Jan...	O001	3000
C009	Monday, Janu...	O112	3000
C001	Thursday, Feb...	O173	3000

To table

DimDate

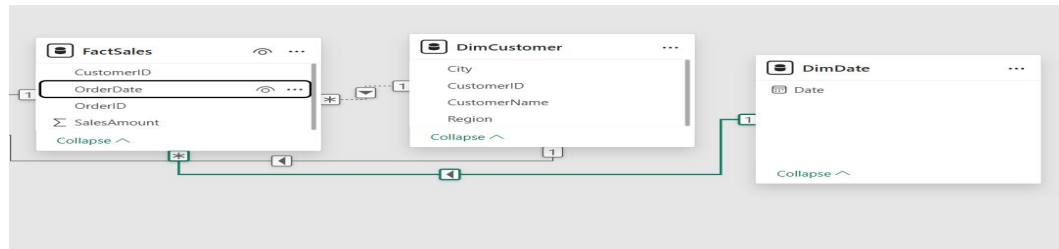
Date
1/1/2025 12:0...
1/2/2025 12:0...
1/3/2025 12:0...

Cardinality

Many to one (*:1)

Cross-filter direction

Single



Queries [6]

MessySales
Customers
Orders
DimCustomer
FactSales
DimRegion

Region

Region
1 Mindanao
2 Visayas
3 Luzon

Table.Distinct("#Removed other Columns")

Structure

Manage relationships
Relationships

New visual calculation
measure

New measure
measure

Quick measure
column

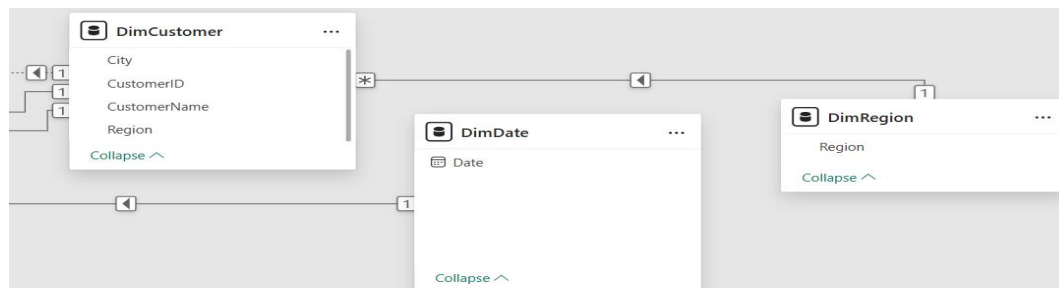
New table
table

Mark as table
table

Calenc

1 DimDate = CALENDAR[DATE(2025,1,1), DATE(2025,12,31)]

CustomerID	Region	SalesAmount
Customer 1	Luzon	5200
Customer 10	Mindanao	1200
Customer 100	Mindanao	16500
Customer 101	Mindanao	6000
Customer 102	Luzon	90800
Customer 103	Mindanao	4200
Customer 104	Visayas	3700
Customer 105	Mindanao	1200
Customer 106	Visayas	52500
Customer 107	Mindanao	9000
Customer 108	Mindanao	46200



Flat Table vs Star Schema

Flat Table

- Single denormalized table
- Unclear chart labels ('Count of Region')
- Filtering is slower
- No proper dimensional relationships

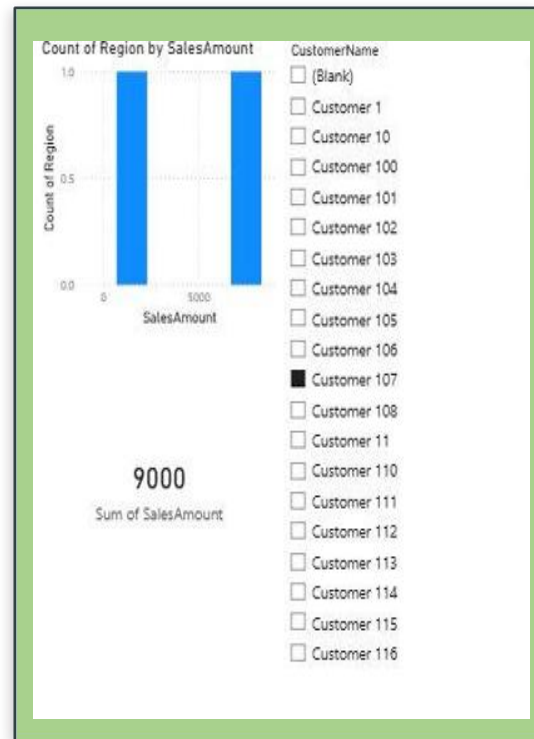
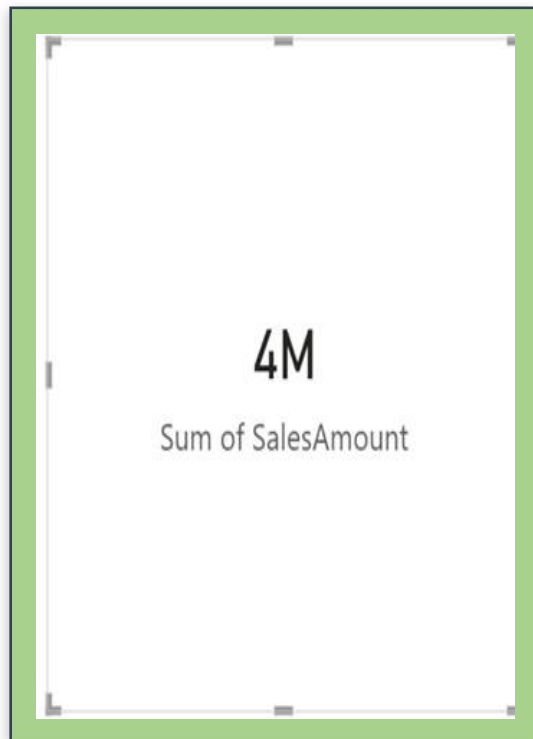
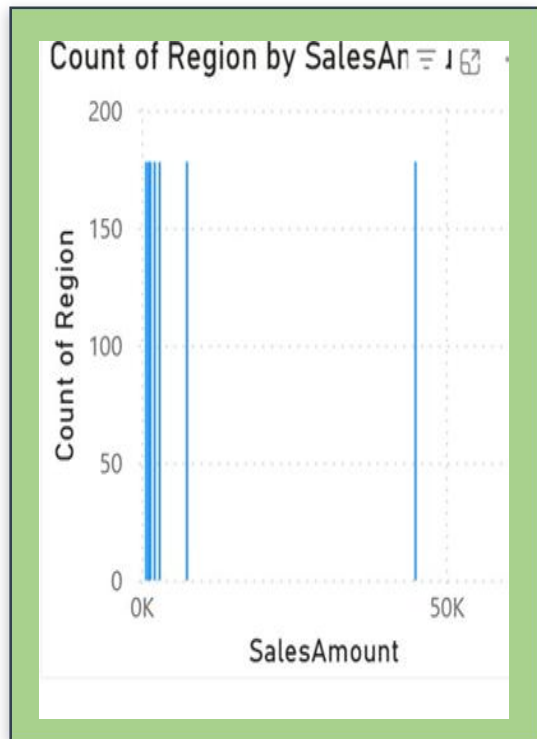
Star Schema

- FactSales linked to DimCustomer & DimDate
- Proper labels ('Sum of SalesAmount by Region')
- Fast cross-table filtering
- DimCustomer slicer filters FactSales instantly

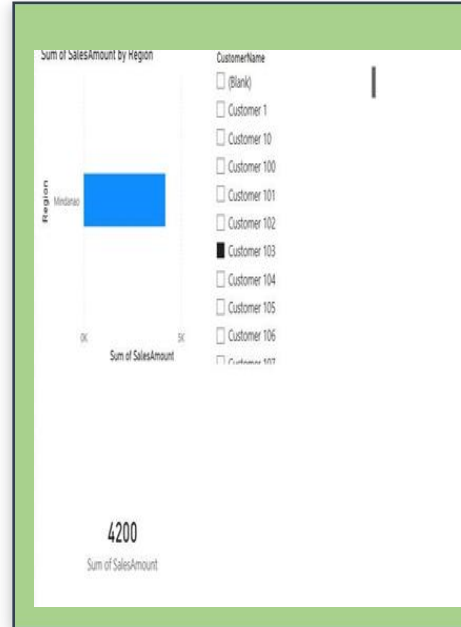
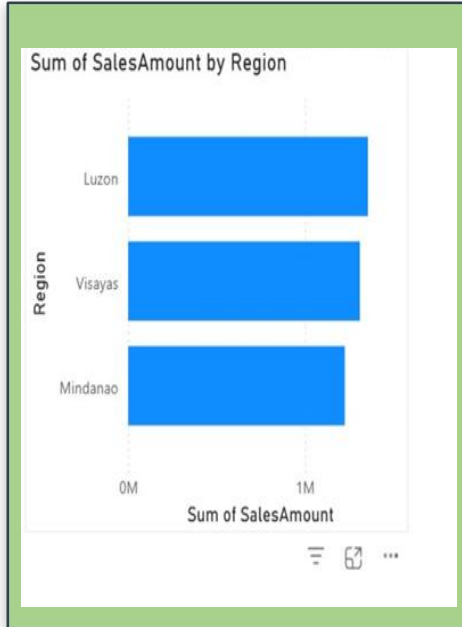
Performance

- Bar chart: 103 ms
- Card visual: 88 ms
- Slicer: 97 ms
- Total load time: 288 ms

Flat Table



Star Schema

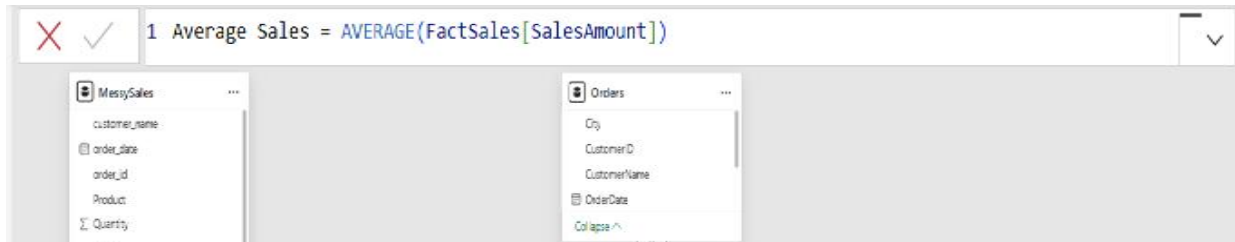
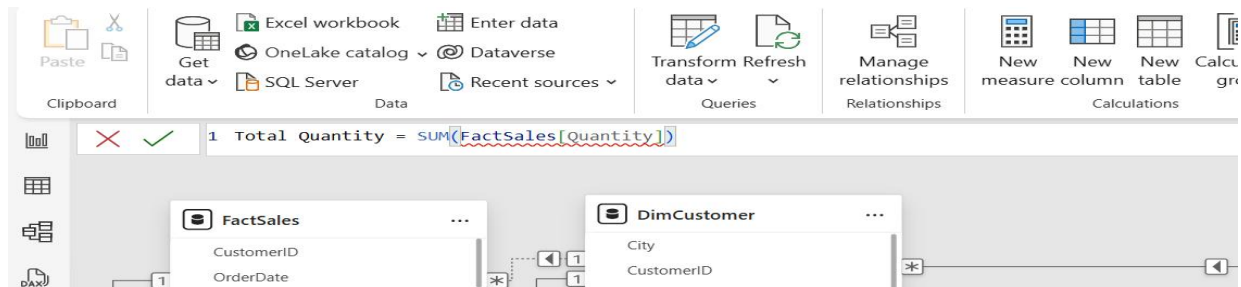
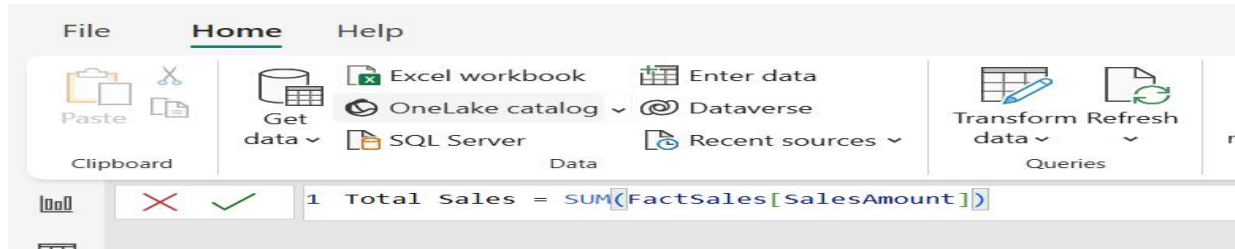


INTRODUCTION TO DAX

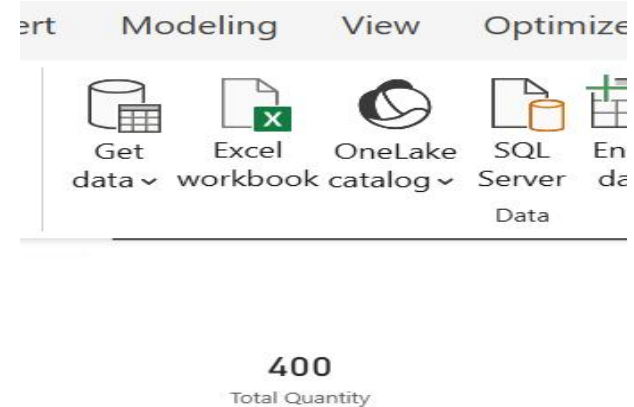
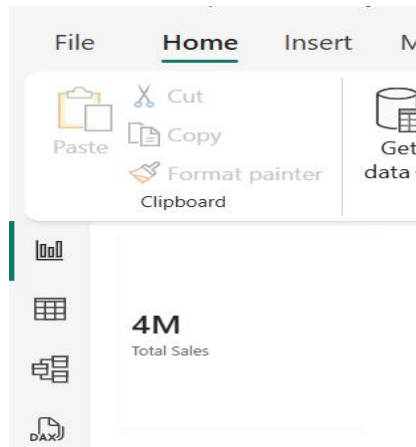
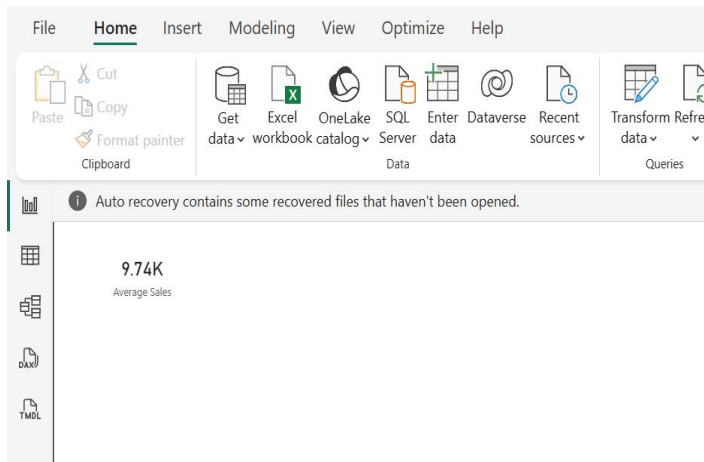
DAX Measures

Create, apply & display basic DAX measures

Enhancement Activities

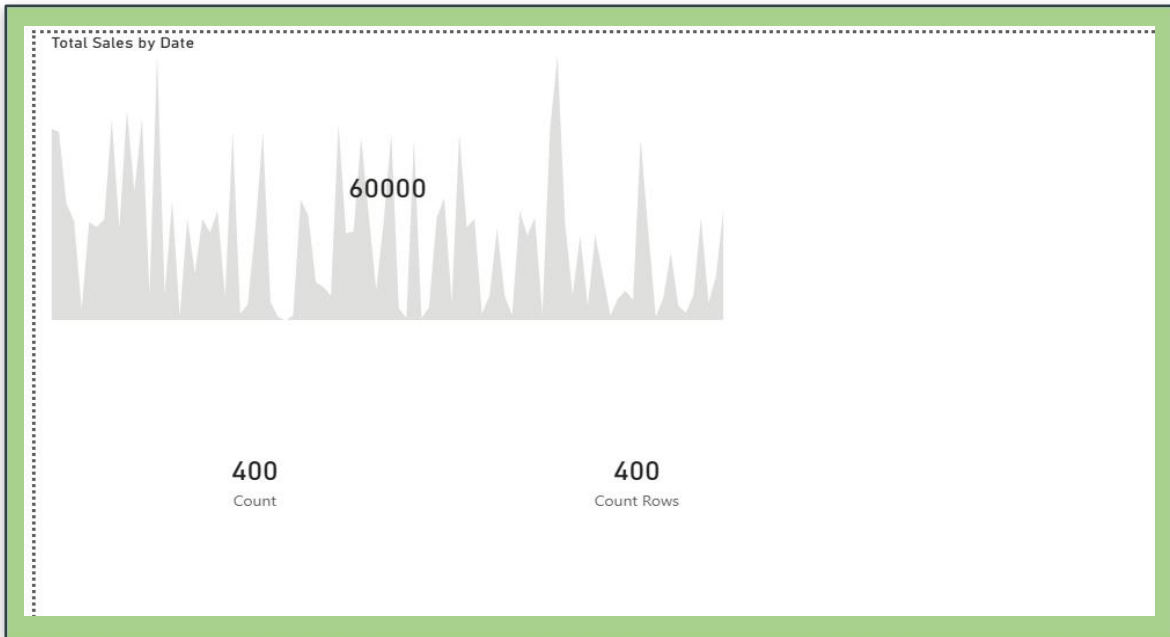


Enhancement Activities



1. What is a DAX measure? A calculation that updates dynamically depending on the current filter context, meaning it recalculates whenever you interact with slicers or visuals.
2. Difference between SUM and AVERAGE? SUM adds up all values to give a total (e.g., Total Sales = 3,894,400), while AVERAGE computes the mean by dividing the total by the number of rows (e.g., 9,736 per order). Use SUM for overall totals and AVERAGE for analyzing values per item or transaction.
3. Why use measures instead of calculated columns? Measures respond to the current context and do not take up storage space. In contrast, calculated columns are fixed values computed for each row and stored in the model, which increases file size.

Enhancement Activity



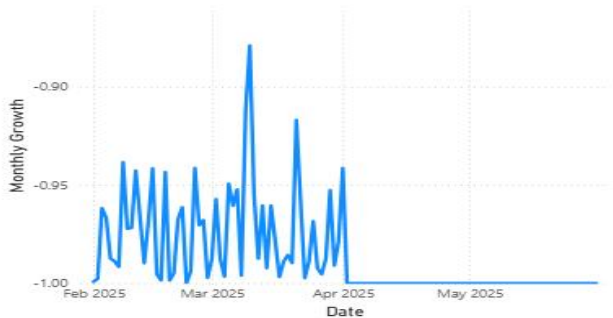
Enhancement Activities

- Create a KPI visual using Total Sales.
- Compare measure vs calculated column performance.
- Explore COUNT vs COUNTROWS.

Enhancement Activities

```
YTD Sales = TOTALYTD(SUM(FactSales[SalesAmount]), DimDate[Date])
```

```
Monthly Growth =  
DIVIDE(  
    [Total Sales] - CALCULATE([Total Sales], PREVIOUSMONTH(DimDate[Date])),  
    CALCULATE([Total Sales], PREVIOUSMONTH(DimDate[Date]))  
)
```



4M
YTD Sales

GUIDE QUESTIONS

Why is a date table required?

Acts as a complete and continuous calendar table for 2025. Without it, time intelligence functions like TOTALYTD and PREVIOUSMONTH cannot properly navigate across periods, often leading to blank or inaccurate results.

How does PREVIOUSMONTH help analyze trends?

Adjusts the filter context to the prior month, enabling Month-over-Month (MoM) calculations such as:
 $\text{Monthly Growth} = \text{DIVIDE}([\text{Total Sales}] - [\text{Prior Month}], [\text{Prior Month}])$.

What is time intelligence in DAX?

Functions like TOTALYTD, PREVIOUSMONTH, and SAMEPERIODLASTYEAR compute values based on date context. They rely on a continuous and properly structured DimDate table to function accurately.

What insights can YTD Sales provide?

Helps monitor overall performance, track progress toward targets, compare year-over-year results, and identify seasonal patterns or trends. For example, YTD Sales = 4M indicates all transactions for 2025 are being correctly accumulated.

Enhancement Activities

```
MoM Growth =  
DIVIDE(  
    [Total Sales] - CALCULATE([Total Sales], PREVIOUSMONTH(DimDate[Date])),  
    CALCULATE([Total Sales], PREVIOUSMONTH(DimDate[Date]))  
)
```

```
YoY Growth =  
DIVIDE(  
    [Total Sales] - CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DimDate[Date])),  
    CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DimDate[Date]))  
)
```

```
1 QTD Sales = TOTALQTD(SUM(FactSales[SalesAmount]), DimDate[Date])
```

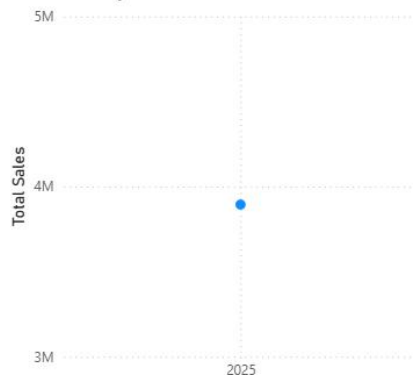
4M

Total Sales

4M

YTD Sales

Total Sales by Year



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QTD Sales